MIL-F-27703D(USAF)
12 May 1970
SUPERSEDING
MIL-F-27703C(USAF)
18 August 1967

MILITARY SPECIFICATION

FILTER-SEPARATOR ELEMENT KIT, LIQUID FUEL FFU-9/E

1. SCOPE

1.1 This specification covers one type of liquid fuel filter-separator element kit, designated FFU-9/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	Plastic Film (Polyethylene Thin Gage)
PPP-P-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

Military

MIL-P-116	Preservation, Methods of
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures For
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 4930

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MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterprcofing, With Appropriate Test Methods
MS9021	Packing, Preformed-AMS 7271, O Ring

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

D256	Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials
D638	Method of Test for Tensile Properties of Plastics
D790	Standard Method of Test For Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Materials

3.3.1 Metallic parts. All metallic parts of the filter-separator element kit shall be made from an aluminum alloy suitable for the application.

3.3.2 Petroleum products. The materials used in the construction of the filter-separator element kit shall neither affect nor be affected by petroleum products.

3.3.3 Fungusproof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

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3.3.4 Plastic parts. If used in the construction of the element, molded plastic parts shall have the following physical properties:

a. Tensile strength minimum . . . . . 4,500 psi

b. Flexural strength minimum . . . . . 7,500 psi

c. IZOD impact (minimum) 1.5 foot pounds per inch of notch

3.4 Design and construction. The filter-separator element kit shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.4.1 Functional design. The filter-separator element kit shall be designed for installation in a Fram Corporation model VFCS-1261-18K2 filter-separator.

3.4.1.1 Coalescer and separator elements. The filter-separator element kit shall consist of 9 coalescer elements and 12 separator elements. The design and dimensions for the coalescer and separator elements shall be as shown on figures 1 and 2. Each coalescer and separator element shall be constructed in one section and shall not consist of several elements cemented together.

3.4.1.1.1 Center tube. The coalescer elements and separator elements shall be provided with an aluminum-alloy center tube.

3.4.1.1.2 End caps. The end caps shall be fabricated from plastic or aluminum suitable for the application and shall be provided with an O-ring groove capable of accepting an MS9021-241 O-ring.

3.4.1.1.3 O-rings. Each coalescer and separator element shall be provided with two MS9021-241 O-rings.

3.4.1.2 Compatible fuels. The filter-separator element kit shall be so designed that it can be used with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.5 Performance. When installed in a Fram Corporation model VFCS-1261-18K2 filter-separator, the filter-separator element kit shall be capable of meeting the performance requirements of MIL-F-8901 with the following exceptions:

a. Removing water to the level specified in MIL-F-8901 at flow rates up to 600 gpm with the influent fuel contaminated with water up to 1 percent by volume.

b. Removing red iron oxide solid contaminant and water up to 1 percent by volume over a period of 125 hours.

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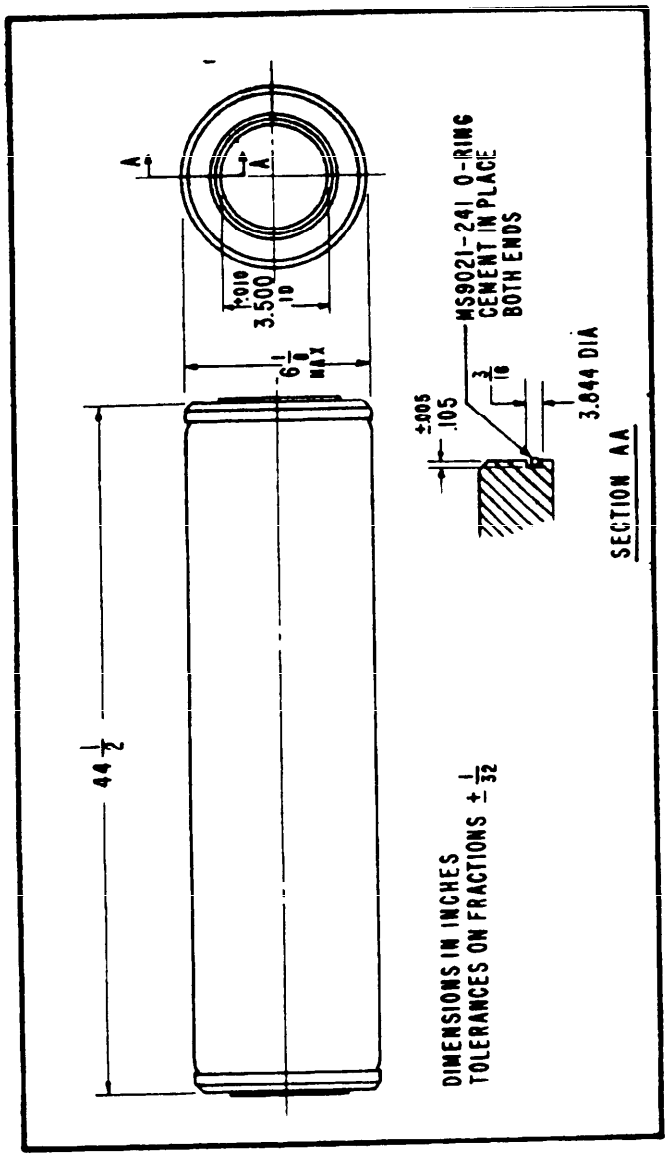


FIGURE 1. Coalescer Element

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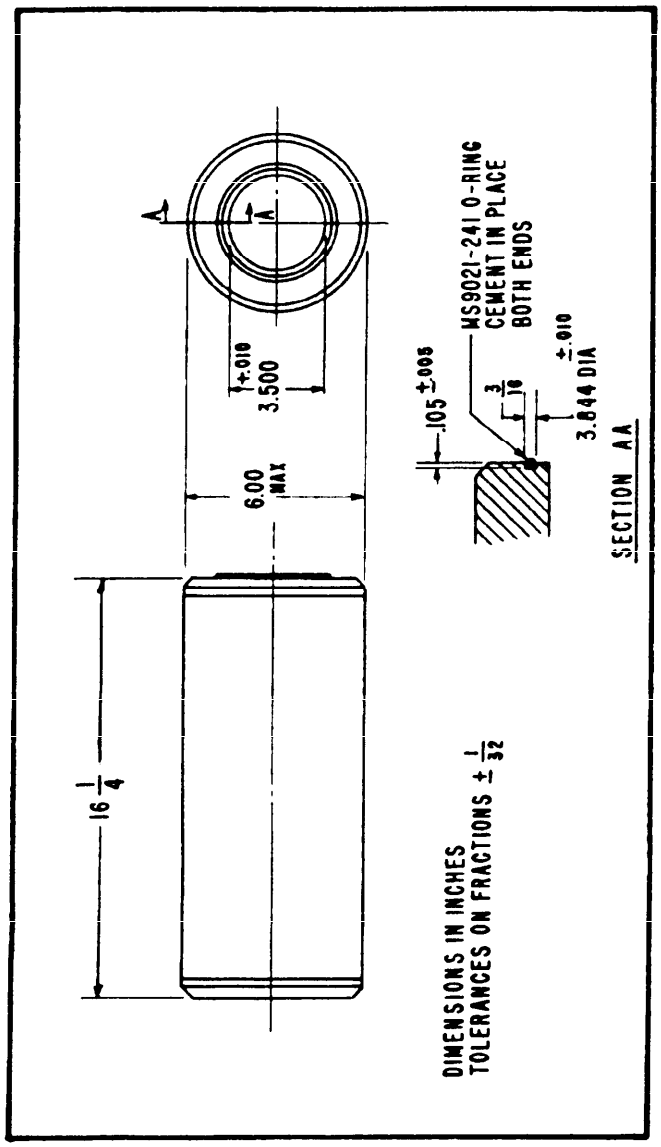


FIGURE 2. Separator Element

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3.6 Identification of product. Only the following information shall be marked on each coalescer and separator element in accordance with MIL-STD-130. If the item is marked on the sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's Part No._____
Manufacturer's Name or Trademark_____
Date of Manufacture_____

3.6.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

3.6.2 Nameplate. Each kit shall be provided with a nameplate to indicate the filter-separator element change criteria. The plate shall be fabricated from type I material conforming to MIL-P-19834 and shall be in accordance with figure 3. Attaching instructions shall be affixed to each nameplate.

3.7 Workmanship. The filter-separator element kit, including all parts, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and testing. The inspection and testing of the filter-separator element kit shall be conducted in a full-scale filter-separator unit in accordance with MIL-F-8901 and as specified herein.

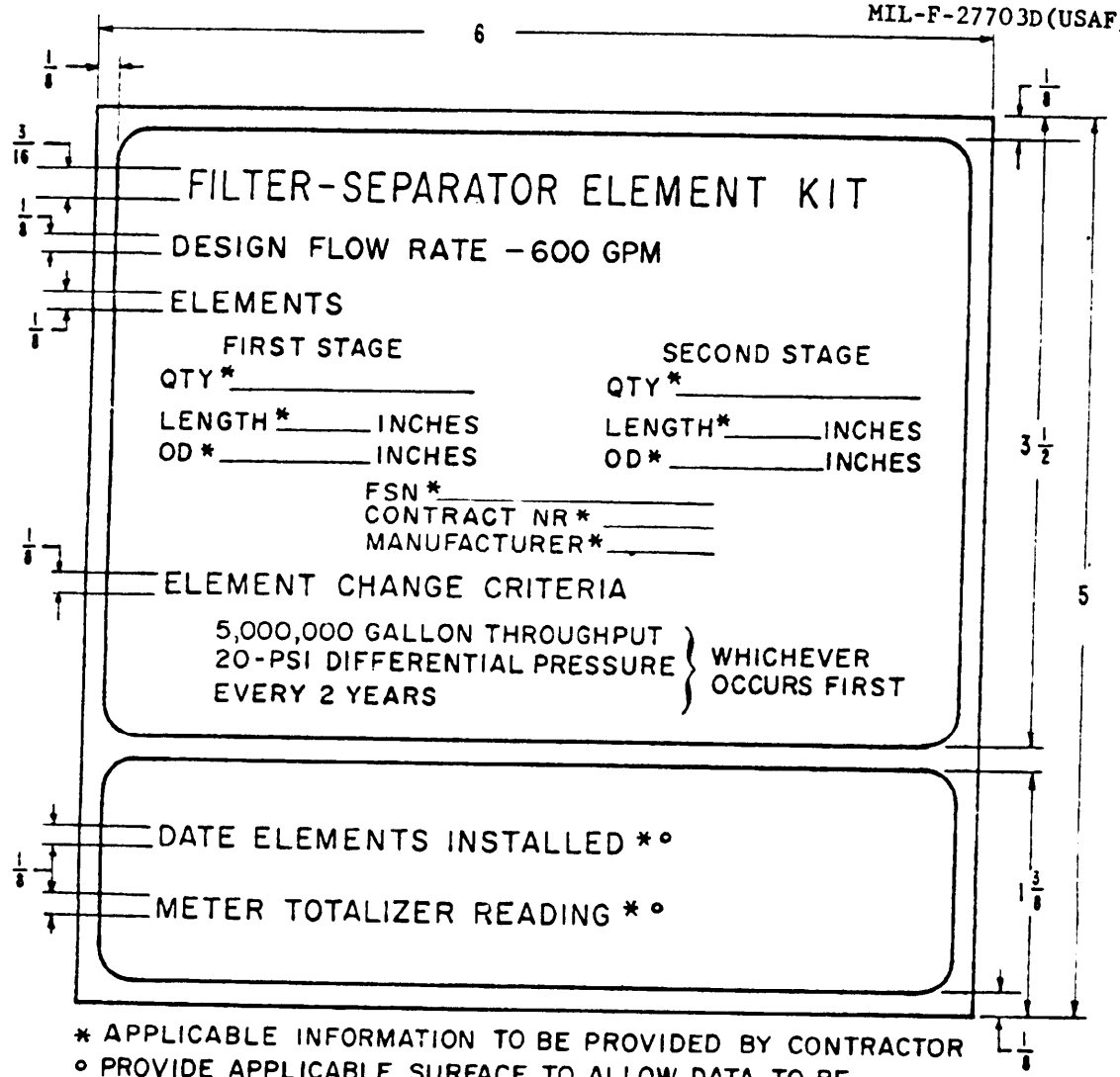
4.1.1 Government inspection. Unless otherwise specified (see 6.2), the Government reserves the right to conduct engineering inspections during the preproduction buildup phase, provide participation during preproduction tests, and perform any inspections after production is released.

4.2 Test condition. During all applicable testing, the filter-separator element kit shall be mounted in a Fram Corporation model VFCS-1261-18K2 filter separator.

4.3 Preproduction testing (see 6.2)

4.3.1 Test samples. Unless otherwise specified (see 6.2), the test samples shall consist of five filter-separator element kits representative of a production lot (see 6.3.1). The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. Samples shall be prepared as specified in 4.3.1.1 and 4.3.1.2.

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* APPLICABLE INFORMATION TO BE PROVIDED BY CONTRACTOR

° PROVIDE APPLICABLE SURFACE TO ALLOW DATA TO BE MARKED BY GOVERNMENT PERSONNEL.

NOTES:

1. ALL LETTERING SHALL BE 3/32-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCES $\pm \frac{1}{32}$

FIGURE 3. Nameplate

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4.3.1.1 Data to accompany test samples. The test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment.

b. Engineering data in the form of assembly drawings (two sets).

4.3.1.2 Test sample and data for the procuring activity. When tests are conducted at a site other than at the laboratory of the procuring activity, the following shall be furnished that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.

b. A sample coalescer and separator element removed from the filter-separator which was subjected to the preproduction tests.

c. Color photographs of the sample coalescer and separator element.

d. A new coalescer and separator element.

4.3.2 Preproduction tests. The preproduction tests shall consist of the tests specified herein and in MIL-F-8901, except as follows:

a. The Water removal tests shall be performed at an input flow rate of 600 gpm. The 5 and 10 percent by volume water removal tests are not applicable to this specification.

b. The Inhibited fuel test may be accomplished using a recirculating-type system.

c. The electric chart recorder shall be used with the Totamitor.

d. During the Red iron oxide and water test, 1 percent water by volume shall be substituted for 3 percent.

e. During the Full scale red iron oxide dry test, when a 40-psi differential pressure across the filter-separator is reached this test shall be temporarily

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suspended. The test shall be continued with a coalescer element removed from the filter-separator and installed in a single-element test fixture until a 75-psi differential pressure is reached. AC test dust or water iron oxide slurry may be used during the single element test as a substitute test contaminant.

f. During the Full-scale test, the flow rate capacity of the filter-separator element kit shall be 600 gpm.

g. Each separator element shall be subjected to the Separator element test, except the test fluid shall be isopropyl-alcohol, Solox 190, chlorothene NU, or other fluids capable of satisfying the test requirements without element damage and which will permit rapid drying of the element prior to packaging for shipment.

h. The coalescer element shall be subjected to the test of MIL-F-8901 entitled Coalescer element.

i. Upon completion of the Life test, the filter-separator housing shall be disassembled and inspected to determine the condition of the filter elements. Defects in the filter element such as swelling of the element, damaged elements, damaged gaskets, et cetera, shall be noted. Excessive swelling of or damage to the element or other parts shall be cause for rejection.

j. Sections of plastic material used in the construction of end caps and elements or especially prepared test specimens shall be tested as indicated below to verify plastic material compliance. These test results shall be included in the test report.

(1) Tensile strength - ASTM D638, type I

(2) Flexural strength - ASTM D790

(3) IZOD impact strength (notched) - ASTM D256, method A.

4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.4.1

b. Sampling plan and test . . . . . See 4.4.2

4.4.1 Individual tests. Each filter-separator element kit shall be examined as required by the Examination of product paragraph of MIL-F-8901, and to determine that it conforms to the filter-separator element kits used in the pre-production tests.

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4.4.2 Sampling plan and test. Samples of the coalescer elements shall be selected at random from each lot in accordance with MIL-STD-105 and subjected to the test specified in 4.3.2h. An AQL of 1.5 percent defective shall be established. The inspection level shall be level I. This test shall be considered destructive.

4.4.3 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be the same for both levels A and C (see 6.2).

5.1.1 Levels A and C

5.1.1.1 Cleaning. The filter-separator element kit shall be cleaned in accordance with process C-1 of MIL-P-116. In addition, all elements shall be air dried and processed in a manner that will eliminate all fire hazards.

5.1.1.2 Unit protection. Each filter element contained in the kit shall be packaged in a polyethylene bag having a minimum thickness of 0.0015 inch. The bags shall be fabricated with material conforming to L-P-378, or equal. The bags shall be sealed to prevent entry of free water, dust, and other foreign matter. Each bagged element shall be packaged in a fiberboard box conforming to PPP-B-636, style RSC, class domestic, grade 200. Boxes shall be sealed with tape conforming to PPP-T-45, or equal.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A and B. Each filter-separator element kit consisting of 9 coalescer and 12 separator elements cleaned and packaged as specified in 5.1 shall be packed for shipment within a close-fitting fiberboard box conforming to PPP-B-640, class 2, grade A, style E. One nameplate as specified in 3.6.2 shall be packaged with each filter element kit. Closure and strapping shall be in accordance with the appendix of PPP-B-640.

5.2.2 Level C. Each filter-separator element kit cleaned and packaged as specified in 5.1 shall be packed one kit to a shipping container. One nameplate as specified in 3.6.2 shall be packaged with each filter element kit. Containers shall meet either the Uniform Freight Classification Rules or rules and regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection. Shipping containers shall be cushioned in accordance with MIL-STD-1186. Free-fall drop tests shall be in accordance with MIL-STD-1186. The tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Filter-Separator Element Kit, Liquid Fuel FFU-9/E.

6. NOTES

6.1 Intended use. The FFU-9/E filter-separator element kit is intended for use as a replacement item in ground aviation liquid fuel filtering and servicing systems. This kit is intended to be installed in a Fram Corporation model VFCS-1261-18K2 filter-separator.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When engineering inspections are not required (see 4.1.1)

c. Locations and conditions for preproduction testing (see 4.3)

d. If other than five kits are required for the preproduction samples (see 4.3.1)

e. Applicable levels of preservation and packaging (see 5.1 and 5.2)

f. Special shipment marking (see 5.4).

6.3 Definition. For the purpose of this specification, the following definition will apply:

6.3.1 Lot. All filter-separator element kits of the same part number manufactured in one production run and submitted for acceptance at the same time will be considered a lot for acceptance purposes.

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6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

AQL . . . . . Acceptable Quality Level

ASTM . . . . . American Society for Testing Materials

FSN . . . . . Federal Stock Number

gpm . . . . . gallons per minute

OD . . . . . outside diameter

psi . . . . . pounds per square inch

QTY . . . . . Quantity

6.5 Contract administration

6.5.1 The contracting organization should require that a prospective contractor shall have or can contract for a suitable test system prior to the award of any contracts for filtration equipment. A test system is the piping, test equipment apparatus, arrangement of test components, laboratory equipment, laboratory supplies, and techniques used during the evaluation of fuel filtration equipment.

6.5.2 The contracting organization should provide for Government inspection during the preproduction buildup and test in addition to providing Government inspection after production is released.

6.5.3 The contracting organization should, prior to issuance of a potential procurement (IFB/REP), contact the engineering activity having a technical responsibility for the specification to confirm that the latest edition of the specification is being cited.

6.6 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4930-F102

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CONTRACT NUMBER

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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